SOC Threat Detection Lab using Splunk: Project Report

1. Introduction

This report details the implementation and outcomes of a Security Operations Center (SOC) threat detection lab utilizing Splunk Enterprise. The project focuses on monitoring, analyzing, and detecting suspicious activities within Windows Security Logs, simulating real-world scenarios such as brute-force attacks. The aim is to provide practical experience in log analysis, threat detection, Security Information and Event Management (SIEM) monitoring, dashboard creation, and alert configuration.

2. Project Overview

The project established a SOC threat detection lab using Splunk Enterprise to simulate and detect suspicious activities, specifically focusing on brute-force attack scenarios. By replicating multiple failed login attempts, the lab demonstrates the workflow of a SOC analyst in detecting, investigating, and responding to security incidents using SIEM tools. This hands-on approach provides valuable experience in critical cybersecurity domains.

3. Objectives

The primary objectives of this project were to:

- Build a basic SOC monitoring environment.
- Collect and analyze Windows Security Event Logs.
- Detect suspicious authentication activity.
- Create security monitoring dashboards.
- Configure automated threat detection alerts.

4. Tools & Technologies

The following tools and technologies were integral to the development and operation of the SOC threat detection lab:

- **Splunk Enterprise**: The core SIEM platform for log ingestion, analysis, and alerting.
- **Splunk Universal Forwarder**: Used for collecting and forwarding Windows Security Event Logs to Splunk Enterprise.
- **Windows 10**: The operating system used to generate security logs and simulate attack scenarios.
- **Virtual Machine Environment**: Provided the isolated infrastructure for setting up the lab components.

5. Lab Architecture

The SOC environment was structured with three main components:

1. **Splunk Server**: Responsible for collecting logs, analyzing security events, and generating alerts.
2. **Windows Machine**: Generates authentication logs and sends them to the Splunk Universal Forwarder.
3. **Splunk Universal Forwarder**: Collects Windows logs from the Windows Machine and forwards them to the Splunk Enterprise SIEM.

The architectural flow can be visualized as follows:

```
Windows Machine
      │
      │  Security Logs
      ▼
Splunk Universal Forwarder
      │
      ▼
Splunk Enterprise (SIEM)
      │
      ├── Log Analysis
      ├── Dashboards
      └── Alerts
```

6. Implementation Steps

The project implementation involved several key steps:

6.1. Lab Environment Setup

- Splunk Enterprise was installed on the main system.
- A Windows machine was set up to generate security logs.
- A virtual lab environment was created for monitoring and analysis.

6.2. Windows Log Collection Setup

- Splunk Universal Forwarder was installed on the Windows system.
- The forwarder was configured to collect Windows Security Event Logs.
- The forwarder was connected to the Splunk server.

6.3. Log Ingestion

- Windows logs were verified to be successfully ingested into Splunk Web.
- A custom index, `index=windows_logs`, was created.
- Security events were confirmed to be visible in Splunk search.

6.4. Suspicious Activity Simulation

- Multiple incorrect login attempts were performed on the Windows system to generate security events mimicking brute-force attack patterns.
- These events were captured and analyzed using Splunk.

7. Log Analysis

Key Windows Event IDs analyzed for threat detection included:

Event ID	Description
4625	Failed Login Attempt
4624	Successful Login
4634	User Logoff
4672	Special Privileges Assigned

An example Splunk Query used to display all failed login attempts is:

```
index=windows_logs EventCode=4625
```

8. Dashboard Creation

SOC monitoring dashboards were developed to visualize security activity, featuring the following panels:

1. Failed Login Attempts
2. Login Activity Timeline
3. Top Users with Failed Logins
4. Authentication Activity Overview

These dashboards are crucial for SOC analysts to quickly identify suspicious patterns.

9. Alert Configuration

An automated alert rule was configured in Splunk Enterprise to detect potential brute-force attacks. The alert condition is triggered when:

More than 5 failed login attempts occur within 1 minute.

Example detection query:

```
index=windows_logs EventCode=4625
| stats count by user
| where count > 5
```

10. Investigation Process

The SOC investigation steps undertaken included:

1. Identifying repeated failed login attempts.
2. Checking user accounts involved.
3. Analyzing login patterns.
4. Investigating abnormal authentication activity.
5. Documenting incident findings.

11. Key Skills Demonstrated

This project demonstrated proficiency in the following key skills:

- SIEM Monitoring
- Log Analysis
- Threat Detection
- Incident Investigation
- Security Dashboard Creation
- Alert Configuration

- SOC Operations

12. Project Outcome

This project successfully provided hands-on experience with Security Operations Center workflows, encompassing:

- Monitoring Windows security logs.
- Detecting suspicious authentication activity.
- Investigating security events.
- Configuring automated alerts.

It effectively simulates a real-world SOC analyst workflow using Splunk Enterprise.

13. Future Improvements

Potential future enhancements for the lab include:

- Integrating additional log sources.
- Implementing advanced threat detection rules.
- Adding automated incident response capabilities.
- Integrating threat intelligence feeds.

14. Author

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